

ABSTRACT

Galactic rotation curves display a well-established empirical regularity: in the outer regions of disk galaxies the rotational velocity tends toward an approximately constant value. In logarithmic form the outer slope

$$s = d \log_{10}(V) / d \log_{10}(R)$$

approaches zero at sufficiently large radii. This behaviour is usually described either through extended dark matter halos or through modified effective dynamical laws. The present work develops an alternative interpretive framework within the Phase–Interference Universe Model (PIUM), in which galaxies are treated as large-scale coherent configurations embedded in a global phase field.

In this description matter corresponds to locally stable, phase-coherent closed configurations, while galactic-scale dynamics is governed by the large-scale organisation and relaxation of the surrounding phase field. The central proposal of the present work is that the outer flat rotation regime reflects the asymptotic relaxation structure of a coherent galactic phase field rather than the cumulative action of independently orbiting discrete mass components alone.

A minimal phase-field formalism is introduced in which the dynamical state of the galaxy is described by a complex field $\Phi = A \exp(i\phi)$, with the phase component carrying the primary large-scale dynamical information. In the inner region local coherent matter configurations dominate the effective dynamics, whereas in the outer region the field approaches a quasi-stationary, scale-free relaxation regime. Under approximate radial symmetry in an effectively disk-like geometry, the outer phase configuration tends toward a logarithmic radial potential, implying an effective radial acceleration scaling as $1/r$ and hence asymptotically flat circular velocity.

The framework is deliberately conservative in its claims. No exact geometric morphology is imposed a priori, no new particles or adjustable halo components are introduced, and no claim of empirical confirmation is made in advance. Instead, the model provides a coherent field-based interpretation linking galactic morphology, large-scale phase organisation, and the emergence of flat rotation curves. A generic morphological coherence observable is proposed as an empirical interface through which future tests may relate outer-slope statistics, velocity scale, and structural organisation in galaxy samples such as SPARC.

This formulation is intended as a field-theoretic interpretive baseline for future confrontation with observational data, including conventional galaxy-dynamical analyses and empirical outer-slope measurements.

1. INTRODUCTION

The outer dynamics of disk galaxies remains one of the central empirical problems of galactic astrophysics. The long-known tendency of observed rotation curves to remain approximately flat at large galactocentric radii stands in tension with the naive expectation that circular velocity should decline once the luminous mass distribution becomes sufficiently centrally concentrated.

In conventional terms, if the outer baryonic mass were the sole dynamical agent and the enclosed mass approached a constant at large radius, one would expect an approximately Keplerian decline,

$$V(R) \sim R^{-1/2}.$$

Instead, many galaxies exhibit an extended outer regime in which the circular velocity remains nearly constant, implying that the effective radial acceleration decreases more slowly than a purely central point-mass field would predict.

Within the standard cosmological interpretation this behaviour is generally attributed to an extended non-luminous halo. Alternative frameworks instead modify the effective dynamical relation in the low-acceleration regime. The present study adopts a different perspective. It does not begin from an independently specified particle halo or from an explicitly modified force law, but from the possibility that the galaxy may be understood as a large-scale coherent configuration embedded in a phase-organised global field.

Within the Phase–Interference Universe Model (PIUM), matter is interpreted not as a set of fundamentally independent point-like constituents, but as the manifestation of stable, locally closed, phase-coherent configurations of a universal wave field. In such a framework it is natural to ask whether the outer dynamics of galaxies may reflect not only the local organisation of matter, but also the asymptotic relaxation structure of the surrounding field itself.

The aim of the present work is therefore to formulate a minimal phase-field description of galactic dynamics in which flat outer rotation curves emerge as the large-scale relaxation signature of a coherent galactic field configuration. The goal is not to claim a complete predictive replacement of existing models, but to provide a mathematically disciplined and physically interpretable route by which flat outer slopes may arise from a global field description.

2. THE GALACTIC PHASE FIELD

Let $\Phi(\mathbf{x},t)$ denote the global wave field describing the dynamical state of the universe. At galactic scales this field may be locally represented by a complex phase field of the form

$$\Phi(\mathbf{r},\theta,t) = A(\mathbf{r},\theta,t) \exp[i\phi(\mathbf{r},\theta,t)],$$

where $A(\mathbf{r},\theta,t)$ is the amplitude envelope and $\phi(\mathbf{r},\theta,t)$ is the phase.

In the present framework the phase component carries the primary large-scale dynamical information. Localised matter configurations correspond to regions where the field forms persistent closed structures of phase coherence. These local closures are interpreted as the material subsystems of the galaxy: stars, gas structures, stellar concentrations, and related bound configurations.

The galaxy as a whole is not therefore described as merely the sum of many independent local objects. Rather, it is treated as a macroscopic coherent organisation of the phase field in which local matter configurations are embedded. This distinction is crucial. In the inner regions local coherent structures may dominate the effective dynamics, whereas at large radii the individual details of those structures may become dynamically secondary relative to the large-scale relaxation pattern of the global field.

The guiding ontological statement is therefore the following:

A disk galaxy is interpreted not primarily as a collection of independently orbiting bodies, but as a globally coherent phase-organised configuration in which local matter loops are embedded within a larger relaxation field.

3. LOCAL STRUCTURE AND GLOBAL RELAXATION

The effective dynamical state of the galaxy is assumed to contain two conceptually distinct contributions.

First, there is a local contribution generated by the coherent matter structures themselves. This component reflects the phase tension associated with the spatial organisation of the bound material configurations.

Second, there is a global contribution associated with the tendency of the surrounding field to relax toward a large-scale equilibrium relative to the null-phase reference state.

Accordingly, define the effective phase potential as

$$\Psi(\mathbf{r}) = \Psi_{\text{loc}}(\mathbf{r}) + \Psi_{\text{null}}(\mathbf{r}),$$

where:

$\Psi_{\text{loc}}(\mathbf{r})$ = local phase tension generated by coherent matter structures,
 $\Psi_{\text{null}}(\mathbf{r})$ = large-scale relaxation potential of the surrounding field.

The effective observed radial acceleration is then written as

$$\begin{aligned} g_{\text{obs}}(\mathbf{r}) &= -d\Psi(\mathbf{r})/dr \\ &= -d/dr [\Psi_{\text{loc}}(\mathbf{r}) + \Psi_{\text{null}}(\mathbf{r})]. \end{aligned}$$

This formulation is not yet a full microphysical derivation. It is an effective large-scale description, analogous in spirit to the use of an effective gravitational potential in standard galactic dynamics. Its significance lies in the separation of regimes.

Inner regime:
 $\Psi_{\text{loc}} \gg \Psi_{\text{null}}$

Outer regime:
 $\Psi_{\text{null}} \approx \Psi_{\text{loc}}$

In the inner galaxy the dynamics is mainly determined by the local organisation of coherent matter loops. In the outer galaxy the local contribution weakens and the large-scale relaxation structure of the field becomes progressively dominant.

This regime split is the conceptual heart of the model. It allows the outer rotation curve to be interpreted not as a direct sum of many local orbital influences, but as the asymptotic signature of the global field configuration.

4. MINIMAL PHASE DYNAMICS

To provide a formal basis for the phase-field description, consider a minimal dynamical equation for the phase variable $\phi(\mathbf{x},t)$. A simple generic form is

$$\partial^2 \phi / \partial t^2 - c^2 \nabla^2 \phi + \lambda |\nabla \phi|^2 = 0,$$

where c is a characteristic propagation scale and λ measures nonlinear self-coupling between phase gradients.

This equation is not proposed here as a final fundamental law. Rather, it serves as a minimal phenomenological representation of three ingredients that a PIUM-like galactic field should plausibly contain:

1. propagation of phase disturbances,
2. spatial relaxation through a Laplacian smoothing term,
3. nonlinear interaction of phase gradients.

Its interpretive usefulness lies not in the full nonlinear solution, but in its asymptotic behaviour.

In the outer regions of a disk galaxy, where the field is assumed to approach a quasi-stationary large-scale state, two simplifications become physically plausible:

1. time-dependent oscillatory contributions become subdominant,
2. nonlinear local gradient interactions weaken relative to the large-scale relaxation pattern.

Under those conditions the outer phase field approaches an approximately source-free relaxation regime,

$$\nabla^2 \phi \approx 0.$$

This is the key asymptotic reduction. It says that sufficiently far from the strongly structured inner region, the phase field is governed mainly by smooth, large-scale relaxation rather than by rapid local forcing.

5. ASYMPTOTIC FIELD STRUCTURE AND THE LOGARITHMIC POTENTIAL

The asymptotic equation

$$\nabla^2 \phi \approx 0$$

has solutions whose radial form depends on geometry. In a three-dimensional spherical geometry the familiar asymptotic potential behaves as $1/r$. In an effectively two-dimensional disk-like geometry, however, the radially symmetric harmonic solution is logarithmic:

$$\phi(r) \sim \ln r.$$

This point is crucial. A disk galaxy is not a spherical system in first approximation; it is a flattened, radially extended structure. In the outer galactic plane the large-scale field therefore need not relax spherically. If the relevant asymptotic relaxation is effectively disk-like, then the logarithmic radial dependence appears as the natural scale-free solution.

The large-scale null-phase relaxation potential is therefore identified asymptotically with this logarithmic structure:

$$\Psi_{\text{null}}(r) \sim \ln r.$$

This does not mean that the entire galaxy at all radii must follow a pure logarithmic potential. It means only that in the outer quasi-stationary relaxation regime, the asymptotic large-scale field tends toward a logarithmic form.

Differentiating gives

$$d\Psi_{\text{null}}/dr \sim 1/r,$$

and therefore

$$g_{\text{null}}(r) \sim 1/r.$$

The circular-velocity relation then gives

$$V^2(r)/r = g(r),$$

so if the outer dynamics is dominated by the null-phase relaxation field,

$$V^2(r) \approx r g_{\text{null}}(r) \approx \text{constant}.$$

Hence,

$$V(r) \rightarrow V_{\text{flat}}$$

at sufficiently large radius.

This is the central result. Flat outer rotation curves emerge not from an assumed halo geometry, but from the asymptotic scale-free relaxation structure of a coherent outer phase field in an effectively disk-like geometry.

6. INTERPRETATION OF THE FLAT REGIME

Within this framework, the flat outer rotation regime is interpreted as the kinematic signature of global field coherence.

The inner galaxy remains structurally complex. Local closed configurations, varying baryonic distributions, and internal phase tensions produce heterogeneous dynamics. The outer galaxy, however, gradually loses sensitivity to the fine-grained internal details of these local structures. The system moves toward a simpler, more global, more weakly scale-dependent state.

The flat rotation regime therefore does not imply that every individual galaxy becomes exactly identical in its outer profile. It implies that the outer ensemble behaviour of a coherent galactic field tends toward a preferred asymptotic state in which the logarithmic slope

$$s = d \log_{10}(V_{\text{obs}}) / d \log_{10}(R)$$

approaches zero.

This gives a natural field-based reading of the empirical fact that flatness is best understood statistically and asymptotically rather than as an exact point-by-point identity.

7. MORPHOLOGICAL COHERENCE WITHOUT ASSUMING A FIXED GEOMETRY

A key point of the present approach is that no precise geometric morphology is imposed a priori. The model does not require the galaxy to be explicitly spiral in a strict mathematical sense, nor toroidal, nor describable by any single rigid large-scale shape class.

Instead, the observable morphology of a disk galaxy is interpreted more generally as the visible trace of large-scale phase organisation.

To quantify this, define a generic morphological coherence parameter κ as a measure of how strongly the observed surface-brightness distribution is dominated by low-order, large-scale azimuthal organisation relative to higher-order or irregular structure. Let

$$I(\mathbf{r},\theta)$$

denote the surface brightness and expand it in azimuthal Fourier modes. Then κ may be defined schematically as the ratio of integrated power in the dominant low-order coherent modes to the total non-axisymmetric power, for example

$$\kappa = [\Sigma_{\text{m}} (\text{m in low-order set}) \int A_{\text{m}}(\mathbf{r}) d\mathbf{r}] / [\Sigma_{\text{m}} (\text{m} \geq 1) \int A_{\text{m}}(\mathbf{r}) d\mathbf{r}],$$

with $0 \leq \kappa \leq 1$.

This definition intentionally avoids committing the framework to one exact morphology. A galaxy with strong coherent low-order organisation will have larger κ . A more irregular or fragmented morphology will have smaller κ .

In this way, morphology enters the model not as a predetermined geometric form, but as an observable measure of the degree of large-scale phase organisation.

8. COHERENCE–DYNAMICS COUPLING

If outer galactic dynamics reflects the large-scale organisation of the phase field, then one should expect some statistical relation between outer slope and morphological coherence.

At the empirical level, the simplest hypothesis is that the logarithmic outer slope depends jointly on the galaxy's dynamical scale and its degree of morphological organisation. A phenomenological relation may therefore be written as

$$s = a - b \log_{10}(V_{\text{flat}}) - c \kappa,$$

where:

V_{flat} = asymptotic rotation-velocity scale,
 κ = morphological coherence parameter,
 a, b, c = empirical coefficients.

This is not introduced as a first-principles law. It is an empirical interface between the field interpretation and observable galaxy structure.

Its qualitative meaning is straightforward. Larger V_{flat} traces larger dynamical scale, while larger κ traces stronger large-scale organisation. If the outer flat regime indeed corresponds to a coherence-dominated relaxation state, then stronger coherence should statistically push the outer slope toward the asymptotic near-zero state.

Thus the expected sign is

$$c > 0.$$

In words: increasing morphological coherence should flatten the outer slope.

This relation provides a direct observational test. The theory is not forced to specify a rigid geometry in advance. It is sufficient that greater large-scale structural organisation, however morphologically realised, should correlate with movement toward the asymptotic flat regime.

9. CHARACTERISTIC TRANSITION SCALE

The model also suggests a natural transition between the inner locally structured regime and the outer globally relaxed regime.

Let the transition occur near the radius where

$$\Psi_{\text{loc}}(r) \approx \Psi_{\text{null}}(r).$$

At this point the outer field ceases to be dominated primarily by local coherent matter tensions and becomes increasingly controlled by the large-scale null-phase relaxation structure.

The corresponding acceleration scale may be written schematically as

$$a_0 \sim |d\Psi_{\text{null}}/dr|_{\text{(transition)}}.$$

This is not introduced here as a universal law constant. Rather, it represents the characteristic scale at which the galaxy shifts from local structure dominance to global field relaxation dominance.

This notion creates a natural bridge to the empirical fact that outer dynamical regularities often appear below a characteristic acceleration scale.

10. EMPIRICAL TEST STRUCTURE

The framework yields several observable consequences.

1. Outer-slope central tendency
If the outer phase field approaches a scale-free relaxation regime, then the outer-slope distribution of disk galaxies should cluster statistically around $s \approx 0$ at sufficiently large radii.

2. Systematic scale dependence
If the field relaxation state depends on global galactic scale, then s should correlate systematically with V_{flat} .

3. Coherence dependence
If morphology reflects phase organisation, then s should also depend on κ .

4. Outer-definition stability
If the effect is genuinely asymptotic, then moving the outer-region definition farther outward should drive the mean or median slope toward zero.

These predictions are directly testable using image decomposition and rotation-curve samples such as SPARC.

11. DISCUSSION

The present framework makes one central conceptual move: it reinterprets the flat outer rotation regime as the asymptotic relaxation structure of a coherent galactic phase field.

This has several advantages.

First, it avoids the need to impose a precise large-scale morphology before the data warrant one. The galaxy need not be declared spiral in a strict wave-geometric sense, nor toroidal, nor describable by a unique predefined form.

Second, it preserves a clear distinction between local matter structure and global field organisation. The flat outer regime is not attributed to a simple continuation of local baryonic influence, but to the large-scale coherence of the surrounding field.

Third, it naturally explains why flatness should be approached asymptotically and statistically rather than as a sharp universal exact law.

At the same time, the framework remains incomplete in an important sense. The logarithmic asymptotic potential is presently justified as the natural outer solution of a quasi-stationary, scale-free, disk-like relaxation field, but not yet derived as a unique solution of a fully specified nonlinear galactic field theory under realistic boundary conditions. The present formulation should therefore be regarded as a controlled asymptotic field interpretation rather than a complete predictive derivation.

This limitation is acceptable at the present stage provided it is stated explicitly. The framework is distinguished when presented not as a final replacement of standard galactic dynamics, but as a disciplined interpretive hypothesis with clear empirical interfaces and falsifiable morphological consequences.

12. CONCLUSION

Within the Phase–Interference Universe Model, a galaxy may be interpreted as a globally coherent phase-organised configuration embedded in a universal wave field. Local matter structures correspond to stable closed phase configurations, while the outer galactic field approaches a quasi-stationary relaxation state.

Under approximate radial symmetry in an effectively disk-like outer geometry, this relaxation field tends asymptotically toward a logarithmic potential,

$$\Psi_{\text{null}}(r) \sim \ln r,$$

which yields an effective radial acceleration scaling as $1/r$ and therefore asymptotically flat rotation curves.

The model does not require a predetermined exact morphology. Instead, it treats observable galactic structure as the visible manifestation of varying degrees of large-scale field coherence. A generic coherence parameter κ is introduced to connect morphological organisation with outer-slope behaviour and velocity scale through an explicitly testable phenomenological relation.

In this way, the flat outer rotation regime is reinterpreted as the dynamical signature of a globally coherent relaxation field. The resulting framework provides a consistent field-based baseline for future confrontation with galaxy morphology, rotation-curve statistics, and empirical relations derived from datasets such as SPARC.

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